

The integration of Information Retrieval, Machine Learning, and Big Data technologies has transformed modern computing systems. Intelligent systems can now analyze enormous amounts of information, understand user behavior, and provide personalized results in real time. These technologies are widely applied in healthcare, finance, education, cybersecurity, and e-commerce. As Artificial Intelligence continues to evolve, the demand for smart data processing systems will continue to grow, making these fields essential for the future of technology and innovation. learning